

PreClaim-GCT: Self-Supervised Graph-Transformer Pretraining for Incident Outcome Prediction from Healthcare Claims

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Abstract. Administrative claims data provide large-scale coverage of healthcare utilization but pose challenges for predictive modeling due to sparse, high-dimensional coding and limited clinical structure, particularly in psychiatric populations. Existing graph and transformer approaches often rely on fully supervised training or longitudinal sequence modeling, which may not fully exploit within-encounter structure. We propose **PreClaim-GCT**, a graph-transformer framework for encounter-level representation learning using self-supervised masked code modeling. Each inpatient encounter is represented as a heterogeneous graph of diagnosis and medication codes anchored at a central node. The model is pretrained using masked code reconstruction to capture within-encounter co-occurrence patterns without outcome labels, and then fine-tuned for incident post-discharge risk prediction. Across outcomes and horizons, self-supervised pretraining improves performance over strong baselines and the same architecture without pretraining. These results demonstrate that encounter-level self-supervised learning enhances post-discharge risk prediction in administrative claims data.

Keywords: heterogeneous graph · graph transformer · self-supervised learning · administrative claims data · healthcare prediction

1 Introduction

Administrative healthcare claims data provide large-scale coverage of healthcare utilization but pose challenges for machine learning, including sparse high-dimensional coding, limited clinical context, and weak supervision—particularly for predicting conditions following inpatient discharge. These challenges are amplified in psychiatric and comorbid populations, where complex diagnosis and medication patterns complicate representation learning [1,2]. Graph and graph-transformer models improve modeling of structured medical data [3,4], but many approaches rely on fully supervised training or longitudinal sequence modeling [5,6], which may be less effective for claims data with sparse labels [1]. We introduce **PreClaim-GCT** (Fig. 1), a self-supervised graph-transformer

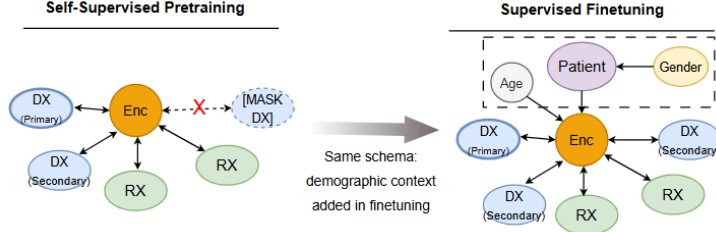


Fig. 1. Encounter-level graph and training phases of PreClaim-GCT.

model for encounter-level representation learning from administrative claims data. Each encounter is represented as a heterogeneous graph of diagnosis and medication nodes anchored at a central node. Unlike prior work, PreClaim-GCT excludes inter-encounter edges to isolate within-encounter structural learning and is pretrained using masked code reconstruction without outcome labels, followed by supervised fine-tuning for incident post-discharge prediction. Empirically, self-supervised pretraining improves performance over strong baselines and generalizes across outcomes with different prevalence levels.

2 Methods

Encounter Graph Construction. Each hospitalization is represented as a heterogeneous graph $G = (V, E)$ (Fig. 1). The graph contains three node types: *Enc* (encounter), *DX* (diagnosis), and *RX* (medication), embedded in a shared space. Primary and secondary diagnoses use separate edge types. During supervised fine-tuning, patient, age, and gender nodes are added for demographic conditioning. A reserved [MASK] token is used during self-supervised pretraining. A subset of diagnosis and medication nodes is masked using [MASK]. Edges connected to masked nodes are removed to prevent identity leakage. Pretraining uses encounter-code relations, while fine-tuning adds patient and demographic edges; no edges connect distinct encounters. **Heterogeneous Graph-Transformer Encoder.** All nodes are embedded in a shared d -dimensional space with relation-specific transformations. Diagnosis and medication embeddings are initialized using hierarchical code structure. At layer ℓ , node representations $\mathbf{h}_v^{(\ell)}$ are updated via relation-aware attention over neighbors $\mathcal{N}_r(v)$ with $H = 2$ heads. Query and key projections are $\mathbf{q}_v^{(k)} = Q_k \mathbf{h}_v^{(\ell)}$ and $\mathbf{k}_u^{(k)} = K_k \mathbf{h}_u^{(\ell)}$, with attention score $e_{uv}^{(k,r)} = \frac{(\mathbf{q}_v^{(k)})^\top \mathbf{k}_u^{(k)}}{\sqrt{d_k}}$. Attention is normalized within each relation:

$$\alpha_{uv}^{(k,r)} = \frac{\exp(e_{uv}^{(k,r)})}{\sum_{u' \in \mathcal{N}_r(v)} \exp(e_{u'v}^{(k,r)})}. \quad (1)$$

Messages are transformed as $\mathbf{m}_{u \rightarrow v}^{(r)} = W_r \mathbf{h}_u^{(\ell)}$, and node updates are

$$\mathbf{h}_v^{(\ell+1)} = \text{LayerNorm} \left(\mathbf{h}_v^{(\ell)} + \sum_r \sum_{k=1}^H \sum_{u \in \mathcal{N}_r(v)} \alpha_{uv}^{(k,r)} V_k \mathbf{m}_{u \rightarrow v}^{(r)} \right). \quad (2)$$

The final encounter-node representation is used for downstream prediction.

Masked Code Modeling with In-Batch Contrastive Loss. PreClaim-GCT is pretrained using masked code modeling on heterogeneous encounter graphs. Let $\mathcal{C}$ denote diagnosis and medication nodes and let $\mathcal{M} \subset \mathcal{C}$ be the masked subset sampled with masking rate ρ . For each masked encounter-code pair (v_k, c_k) , a positive pair is formed between encounter embedding $\mathbf{h}_{v_k}$ and code embedding $\mathbf{e}_{c_k}$. All other masked codes in the minibatch serve as implicit negatives. We optimize an in-batch InfoNCE objective [8]:

$$\mathcal{L}_{\text{SSL}} = - \sum_{k=1}^M \log \frac{\exp(\text{sim}(\mathbf{h}_{v_k}, \mathbf{e}_{c_k})/\tau)}{\sum_{j=1}^M \exp(\text{sim}(\mathbf{h}_{v_k}, \mathbf{e}_{c_j})/\tau)}, \quad (3)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and $\tau = 0.07$ is the temperature parameter. The pretrained encoder is later fine-tuned end-to-end for supervised incident outcome prediction.

3 Downstream Tasks and Experimental Setup

Study Cohort and Prediction Setting. We used New York State (NYS) Medicaid claims data (2016–2023), including individuals with at least one psychiatric hospitalization during 2016–2021. The cohort comprised approximately 159,000 patients and 759,000 inpatient encounters (16.9 records per encounter). Data were split at the patient level into training (70%), validation (15%), and test (15%) sets. All training data were used for self-supervised pretraining, while 20% of training patients were sampled for supervised fine-tuning and resampled five times. Continuous enrollment was required for 365 days prior to admission and 365 or 730 days post-discharge. Each prediction instance corresponds to an index encounter represented as the heterogeneous graph in §2. **Incident Post-discharge Outcomes.** We evaluate two tasks using 365- and 730-day windows: **new-onset diabetes** and **new-onset depression**. Incident outcomes are defined as the first diagnosis within the post-discharge window among individuals without prior history; remaining patients are labeled as positive or negative cases. **Baseline Models and Evaluation Setup.** We compare against logistic regression (LR), XGBoost, graph neural networks (GCN [7], GAT [9]), a BEHRT-style transformer [5], and two variants of PreClaim-GCT. All models use identical patient-level splits and input features to ensure fair comparison and are evaluated using AUC-ROC and AUC-PR. LR and XGBoost use multi-hot features; GCN and GAT use the same encounter graph; and BEHRT uses diagnosis sequences without explicit graph edges. PreClaim-GCT (Random) and (SSL) share the same architecture, isolating the effect of self-supervised pretraining. **Training Details.** Models were trained using Adam with hidden dimension 128, three layers, and two attention heads. During pretraining, 15% of nodes are masked and reconstructed using an in-batch InfoNCE objective with temperature $\tau = 0.07$, following common practice [8]. The pretrained encoder is then fine-tuned end-to-end with early stopping based on validation AUC-PR.

Table 1. Incident outcome prediction (mean $\pm$ std across five resampled splits). Best performance in each column is shown in bold.

Model	Diabetes		Depression	
	AUC-ROC	AUC-PR	AUC-ROC	AUC-PR
365 days				
LR	0.632 $\pm$ 0.017	0.110 $\pm$ 0.006	0.589 $\pm$ 0.008	0.285 $\pm$ 0.007
XGBoost	0.673 $\pm$ 0.010	0.130 $\pm$ 0.003	0.618 $\pm$ 0.009	0.313 $\pm$ 0.007
GCN [7]	0.675 $\pm$ 0.011	0.121 $\pm$ 0.007	0.597 $\pm$ 0.005	0.284 $\pm$ 0.009
GAT [9]	0.662 $\pm$ 0.021	0.116 $\pm$ 0.010	0.583 $\pm$ 0.006	0.275 $\pm$ 0.006
BEHRT [5]	0.678 $\pm$ 0.009	0.121 $\pm$ 0.006	0.606 $\pm$ 0.013	0.297 $\pm$ 0.011
PreClaim-GCT (Random)	0.677 $\pm$ 0.004	0.118 $\pm$ 0.005	0.623 $\pm$ 0.008	0.309 $\pm$ 0.005
PreClaim-GCT (SSL)	0.690$\pm$0.002	0.134$\pm$0.003	0.633$\pm$0.004	0.322$\pm$0.005
730 days				
LR	0.641 $\pm$ 0.019	0.186 $\pm$ 0.010	0.596 $\pm$ 0.006	0.399 $\pm$ 0.005
XGBoost	0.675 $\pm$ 0.011	0.208 $\pm$ 0.004	0.627 $\pm$ 0.009	0.432 $\pm$ 0.012
GCN [7]	0.670 $\pm$ 0.007	0.197 $\pm$ 0.005	0.596 $\pm$ 0.022	0.389 $\pm$ 0.022
GAT [9]	0.650 $\pm$ 0.014	0.178 $\pm$ 0.011	0.586 $\pm$ 0.027	0.380 $\pm$ 0.024
BEHRT [5]	0.681 $\pm$ 0.003	0.206 $\pm$ 0.008	0.614 $\pm$ 0.014	0.415 $\pm$ 0.015
PreClaim-GCT (Random)	0.671 $\pm$ 0.009	0.197 $\pm$ 0.009	0.619 $\pm$ 0.001	0.423 $\pm$ 0.006
PreClaim-GCT (SSL)	0.686$\pm$0.003	0.215$\pm$0.004	0.634$\pm$0.014	0.444$\pm$0.007

4 Results

Table 1 reports incident prediction performance. PreClaim-GCT achieves the best results. For diabetes, pretraining improves ROC by $\sim 2\%$ and AUC-PR by 10–15% over the randomly initialized model, with similar gains observed for depression. These trends persist at the 730-day horizon, where PreClaim-GCT (SSL) continues to outperform all baselines, including XGBoost, BEHRT, and the same architecture without pretraining. **Statistical Significance Analysis.** We assessed statistical significance following [10] using a Friedman test across all seven models based on AUC-PR. The analysis includes five repeated subtraining splits and four prediction settings, yielding 20 evaluation blocks. The Friedman test indicated significant differences across models ($\chi^2 = 74.16$, $p < 0.001$). PreClaim-GCT (SSL) achieved the best average rank (1.13), followed by XGBoost (2.09) and BEHRT (3.75). Post-hoc Wilcoxon signed-rank tests with Holm correction confirmed that PreClaim-GCT (SSL) significantly outperformed PreClaim-GCT (Random), XGBoost, and BEHRT ($p < 0.001$).

5 Discussion and Conclusion

Interpretation of Representation Gains. Larger improvements for depression suggest that masked code modeling captures dense co-occurrence patterns, whereas diabetes prediction may depend more on longer-term processes. Consistent gains across outcomes indicate improved representation quality. Although

gains over strong baselines are modest, comparison between PreClaim-GCT (Random) and (SSL) isolates the effect of self-supervised pretraining. Because both models share the same encoder, improvements arise from representation learning. In claims data with sparse labels, even modest AUC-PR gains remain meaningful. **Implications for Claims-Based Modeling.** The framework leverages diagnosis–medication co-occurrence to learn relational structure without curated ontologies, making it suitable for large-scale claims data. Operating at the encounter level, it generalizes across outcomes without requiring longitudinal modeling. **Conclusion and Limitations.** We presented PreClaim-GCT, a graph–transformer model combining self-supervised masked code pretraining with supervised fine-tuning for incident outcome prediction from claims data. The approach consistently outperforms strong baselines, demonstrating the effectiveness of encounter-level self-supervised representation learning. This study focuses on a single state claims system. Hyperparameters were selected based on validation performance and common practice; systematic sensitivity analysis is left for future work. Future work will evaluate generalizability and explore integration with longitudinal modeling.

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